

AMENDMENT NO. 1 MARCH 2015
TO
IS 7098 (PART 2) : 2011 SPECIFICATION FOR CROSSLINKED POLYETHYLENE
INSULATED THERMOPLASTIC SHEATHED CABLES

PART 2 FOR WORKING VOLTAGES FROM 3.3 KV UPTO AND INCLUDING 33 KV

(Second Revision)

(Page 1) — add caption '**SECTION 1 GENERAL**' before scope.

(Page 1, Note 1, under clause. 1.1) — Note 1 to read as 'Cables of 6.35/11kV grade (11 kV earthed system)' are also suitable for use in '6.6 kV/6.6 kV (6.6 kV unearthed system)'.

(Page 1, clause 1.4) — Substitute 'Annex C' for 'Annex A'.

(Page 1, clause 2, line 15) — Substitute '8130 : 2013' for '8130 : 1984'.

[Page 2, (Part 62) : 1993] — Title to read as 'Flame retardance' for 'Fire resistance'

(Page 2) — Insert caption '**SECTION 2 MATERIAL**' before clause 4.

[Page 3, Table 1 — Sl No. iii) (a) Treatment] — duration to read as '7 days' in place of '7 day'.

[Page 3, Table 2 — Sl No. i), col 3, line 5] — substitute 'tested' for 'test'.

(Page 3, clause 7.1 a) — Substitute 'or' for 'and'.

(Page 3, clause 8.1 a) — Substitute 'or' for 'and'.

(Page 3, clause. 8.2, first line) — Substitute 'The galvanized steel wires/formed wires shall conform to IS 3975' for 'The galvanized steel wires/strip shall comply with the requirements of IS 3975'

(Page 3, clause 9) — Please add caption '**SECTION 3 CONSTRUCTION**' below foot note.

(Page 3, clause 10.2) — Substitute 'In case of 1.9/3.3 kV and 3.3/3.3 kV cables, a protective barrier may be applied between the conductor and insulation at the discretion of the manufacturer. Such barrier, when used, shall be compatible with insulating material and suitable for the operating temperature of the cable' for the existing.

(Page 3, clause 11.1, first line) — Substitute '3.3/3.3 kV' for '1.9/3.3 kV'

[Page 4, Table 3, Sl No. ii) a), col 3] — Substitute '10 days' for '7 days'

(Page 4, clause 12.4, title) — Substitute 'Eccentricity' for 'Eccentricity and Ovality'.

(Page 4, clause 13.1, first line) — Substitute '3.3/3.3 kV' for '1.9/3.3 kV'

(Page 4, Table 4, first row, col 8) — Substitute ' - ' for '6.0'

(Page 4, Table 4, first and second row, col 9) — Substitute ' - ' for '8.8'

[Page 5, clause 14.1) — Note given after c)] — Delete and add at the end

'NOTE — It is permissible to apply the strip under metallic part of insulation screen'.

(Page 5, clause 16.3) — Substitute 'The thickness of inner sheath (common covering) is given in Table 5. Single core cables except those specified on '16.2.1' shall have no inner sheath' for the existing.

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(Page 5, clause 17.1.2) — Substitute the 'para' for the existing.

‘The direction of lay of armour shall be left hand. For double wires/formed wires armoured cables, this requirement shall apply to the inner layer of wires/formed wires. The outer layer shall, except in special cases, be applied in the reverse direction to the inner layer and there shall be a separator of suitable non-hygrosopic material; such as plastic tape, bituminized cotton tape, bituminized hessian tape, rubber tape, proofed tape between the inner and outer layers of armoured wires/formed wires.’

(Page 5, clause 17.2) — Insert new para as under :

‘The armoured wires/formed wires shall be applied as closely as possible with coverage not less than 90 percent. The coverage of armour shall be checked as per Annex F’

(Page 5, clause 17.3) — Add the following at the end of existing sentence.

‘The tolerance on nominal dimensions shall be as per IS 3975. However, for formed steel wires compliance shall be ensured only for dimensions ‘A’ and ‘C’.

(Page 6, Table 6) — Replace ‘i) and ii)’ with ‘a) and b)’ and add word” or’ in between option a) & b)

(Page 6, clause 17.5, second line) — Substitute ‘armour’ for ‘armoured’.

[Page 6, clause 18.1 a)] — Substitute ‘Over the non-magnetic metallic tape covering the insulation or over the non-magnetic metallic part of insulation screening in case of unarmoured single core cables, and’ for the existing.

(Page 6, clause 18.3.1, line 4) — Substitute ‘col 4’ for ‘col 3’.

(Page 6, clause 18.3.1, line 5) — Substitute ‘minimum’ for ‘nominal’.

(Page 6, clause 18.3.1, line 6) — Substitute ‘col 5’ for ‘col 4’.

(Page 6, clause 18.3.2, line 2) — Substitute ‘col 6’ for ‘col 5’

(Page 7, clause 19) — Add caption ‘**SECTION 4 TESTS**’ above the clause.

[Page 7, clause 19.1, Sl No. vii) to xv), col 3] — Substitute ‘20.9, 20.2, 20.3, 20.4, 20.5, 20.6, 20.7, 20.8’ for ‘19.9, 19.2, 19.3, 19.4, 19.5, 19.6, 19.7 and 19.8’.

[Page 7, clause 19.1, iii) c] — Delete and renumber the remaining.

[Page 7, clause 19.1, iv), col 2] — Substitute ‘Test for eccentricity of Insulation and thickness of Insulation and sheath’ for existing.

[Page 7, clause 19.1, iv), col 3] — Substitute ‘clause 12’ for ‘clause 11’

[Page 7, clause 19.1, Sl No. vii), col 4] — Substitute ‘20.9’ for ‘19.9’.

(Page 7, clause 19.1.1) — Substitute ‘Tests viii), ix), x), xii) and xiii) are applicable only to screened cables. However, test x) is applicable to cables with rated voltage 6.35/11 kV and above only’ for the existing.

[Page 8, clause 19.1.2, Sl No. f) and g)] — Substitute the following for the existing

- ‘f) Impulse withstand voltage test followed by 15 minutes high voltage test
- g) High voltage test for 4 hours’

(Page 8, clause 19.1.2.1) — Substitute the following for the existing.

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‘If a sample fails in test g), one more sample shall be taken for this test preceded by tests b) and e)’.

[Page 8, clause 19.2, Sl No. e)] — Substitute ‘Test for Eccentricity of insulation and thickness of insulation and sheath’ for the existing

[Page 8, clause 19.2, Sl No. m)] — Delete.

(Page 10, clause 21) — Add caption ‘SECTION 5 IDENTIFICATION, PACKING AND MARKING’ above the clause.

(Page 10, clause 21.1, para 2, line 4) — Substitute ‘Bureau of Indian Standard’ for ‘Indian Standard Insulation’.

(Page 11, ANNEX A) — Substitute ‘(clauses 12.4)’ for ‘(clause 1.4 and 12.4)’

(Page 11, ANNEX A, A-2) — Delete.

(Page 14) — Add Annex F as under —

ANNEX F
(Clause 17.2)
ARMOUR COVERAGE PERCENTAGE

$$\text{Percent coverage} = \frac{N \times d}{W} \times 100$$

where

N = number of parallel wires,
 d = diameter of wire/width of formed wires,

$W = \pi \times D \times \cos a$,

D = diameter under armour,

a = angle between armouring wire/formed wires and axis of cable,

$\tan a = \pi \times D/C$, and

C = lay length of armouring wires/formed wires.

